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Project Title

**EGX360: Intelligent Stock Market
Analysis and Forecasting Platform**

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By

Mohamed Taher Amin

Supervisors

Dr. Samy Abdelnabi

Associate Professor, Faculty of Computer Science
King Salman International University

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ABSTRACT

This thesis presents **EGX 360**, an AI-driven FinTech platform designed to mitigate volatility and data fragmentation in the Egyptian Exchange (EGX). The study demonstrates that integrating macroeconomic features (USD/EGP, gold, interest rates) with a Stacking Ensemble architecture (XGBoost & LightGBM) significantly enhances prediction accuracy. By utilizing Target Engineering (EMA10 direction), the model achieved an **85.22% accuracy and 80% down-trend recall on 28 years of data, yielding a +159% alpha** over the market baseline.

The ecosystem utilizes a scalable AWS and Supabase backend with autonomous scrapers and real-time WebSocket streams. It features an advanced bilingual NLP sentiment pipeline and a cross-platform Flutter frontend built on Feature-First Clean Architecture. Key functionalities include interactive charting, a Cerebras Llama 3.1-powered AI assistant, a paper-trading simulator with a dynamic Guardian Engine for risk management, and a social trading community. Ultimately, EGX 360 provides a production-ready environment for independent investors to achieve data-driven financial autonomy.



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CHAPTER

INTRODUCTION

1.INTRODUCTION

1.1. Introduction

The Egyptian Exchange (EGX) is a cornerstone of the national economy, acting as a primary barometer for investment and corporate growth in the region. As the financial landscape undergoes a rapid digital transformation, the role of Artificial Intelligence (AI) has become indispensable in interpreting market complexities. **EGX 360** is an intelligent, multi-platform financial ecosystem designed to empower individual investors by bridging the gap between raw market data and actionable insights. By integrating advanced quantitative modeling with modern software engineering, the project provides a unified environment for market prediction, automated analysis, and risk-managed trading simulation.

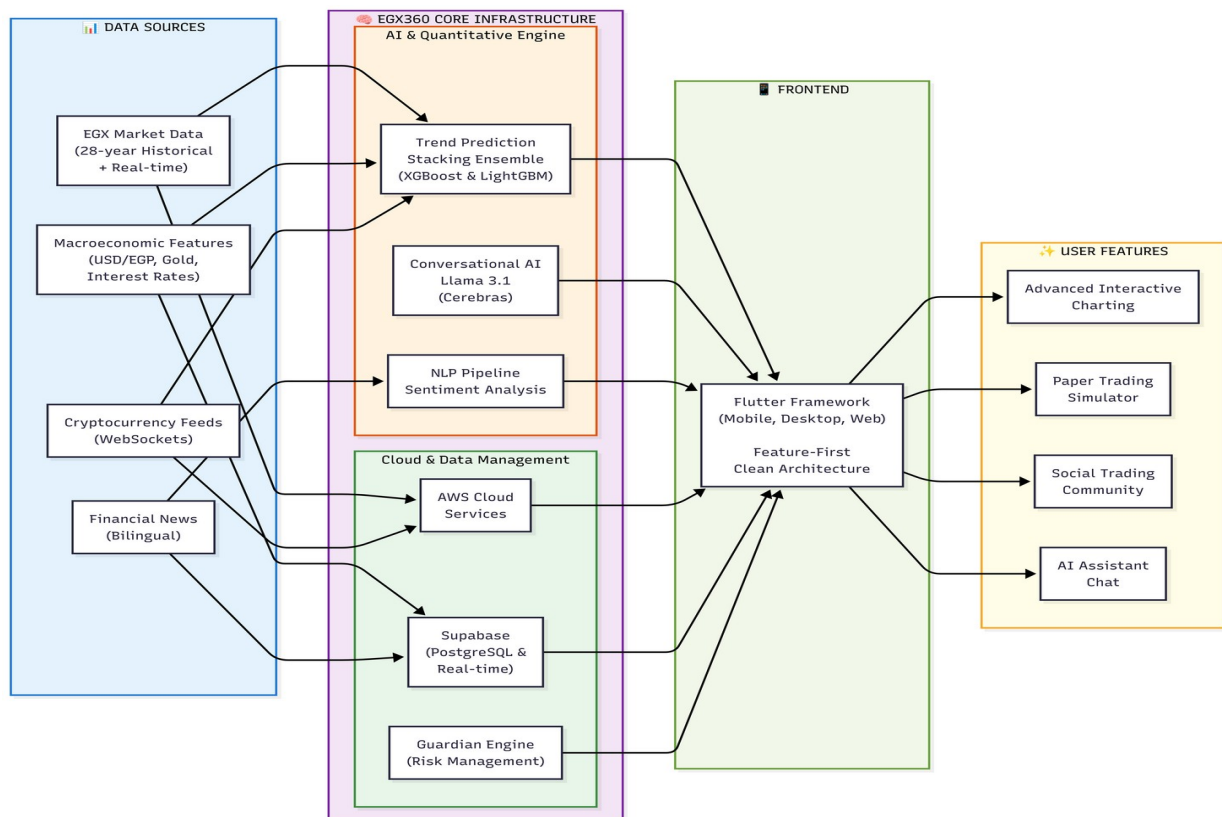


FIGURE 1.1: HIGH-LEVEL CONCEPT OF THE EGX 360 ECOSYSTEM.

1.2. PROBLEM STATEMENT

Despite the availability of market data, independent retail investors in Egypt face significant barriers that hinder effective decision-making:

- **Data Fragmentation and Complexity:** Financial news, technical indicators, and macroeconomic data are often scattered across disparate sources, leading to information overload and analysis paralysis.
- **Market Noise in EGX30:** The Egyptian market is characterized by high volatility and "noise," where traditional analysis methods often fail to distinguish between random fluctuations and significant trend shifts.
- **Lack of Integrated Intelligent Tools:** There is a notable absence of local platforms that combine real-time sentiment analysis, AI-driven conversational guidance, and automated risk management in a single accessible interface.

1.3. OBJECTIVES

The primary goal of this project is to build a robust, production-grade financial intelligence platform that bridges the gap between complex data and individual investors. The specific objectives include:

- **Multimodal Quantitative Forecasting:** Developing a sophisticated predictive engine using a Stacking Ensemble architecture (XGBoost and LightGBM). This objective extends beyond price history to include the integration of macroeconomic features (USD/EGP rates, gold prices, interest rates) and bilingual NLP sentiment analysis (Arabic and English) to achieve high classification accuracy and market alpha.
- **Production-Grade Software Architecture:** Engineering a high-performance, cross-platform application using the Flutter framework. A key objective is implementing a "Feature-First Clean Architecture" to ensure system scalability, maintainability, and a seamless user experience across Web, Android, and Desktop environments.

- **Autonomous Data Orchestration:** Establishing a resilient backend infrastructure on AWS and Supabase. This includes building autonomous data scrapers for the Egyptian Exchange (EGX) and real-time WebSocket streams for cryptocurrency markets to ensure zero-latency data availability.
- **Intelligent Investor Ecosystem:** Integrating advanced AI services, including a real-time financial chatbot powered by the Llama 3.1 model via Cerebras for contextual market support. Additionally, the project aims to deploy a "Guardian Engine" for automated risk management and a real-time social community to foster interactive trading.
- **Risk-Free Financial Simulation:** Creating a high-fidelity "Paper Trading" environment that allows users to test strategies using real market data without financial risk, supported by automated portfolio analytics and performance tracking.

1.4. PROJECT SCOPE

The scope of EGX 360 is defined by the following technical and functional boundaries:

- **Target Markets:** Focuses primarily on the Egyptian Stock Exchange (EGX30 and EGX70 indices) and major Cryptocurrency pairs (e.g., BTC/USD).
- **Data Horizon:** Utilizes over 28 years of historical EGX data, supplemented by real-time macroeconomic features such as USD/EGP exchange rates, gold prices, and national interest rates.
- **Technological Infrastructure:** Deployment on AWS cloud infrastructure, utilizing Supabase for secure backend management and real-time data streaming.
- **Methodology:** Implementation of "Feature-First Clean Architecture" to ensure system scalability and maintainability.

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literature survey

2.literature survey

2.1. survey

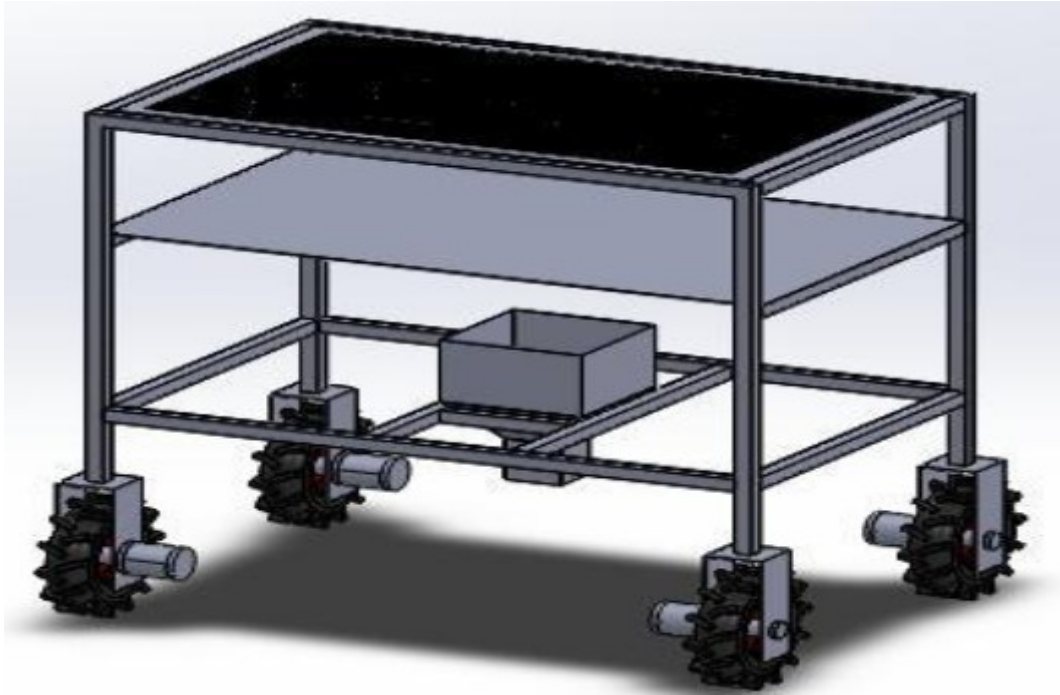


Figure literature survey.1 Solar-powered Agriculture Robot Design [2]

2.2. Design

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3. Mechanical Design

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6.Title of Chapter 6

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Conclusions

7. Conclusions

7.1. Survey

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